

in common robotics research areas. While dealing with robotics, basic functioning can be broadly grouped into location, vision and motion. Advanced techniques for each of the above is what forms the basis of MRPT.

Go the fast and furious way

Ask a robotics engineer and he will tell you about one of the classic robotic problems: the navigation problem. The more creative ones call it the piano mover's problem. Complicated though it sounds, it is not too difficult to comprehend—just imagine trying to make your robot play the *Harry Potter* theme song.

Pause and dissect. Motion planning involves breaking up motion into discrete steps, each satisfying movement constraints and even maybe optimises some aspects of the same. Be it in autonomy, automation, animation, video games or robotic surgery, this is what is done. MRPT uses holonomic algorithms for depicting nearness diagrams and virtual force fields to help set your robot into action.

Plan and connect. Path of the robot is fixed by you via planning algorithms and path planning with occupancy grid maps that iterate to find the shortest route possible. The robot is aware of its path with the reactive navigation mechanism kicking in on encountering an obstacle. All of this can be observed clearly on your computer screen with help from Navigation Viewer, making analysis all the more easy.

You can even export the robot trajectory and point clouds of obstacles to MATLAB for generating high-quality graphs. Metric map can be represented as landmark map, beacon map, coloured point map or even gas concentration map to aid you to convert observations into meaningful data.

Why we need a new library

"Many good scientific programs and programming libraries exist out there. When working with matrices, vectors and graphical representations, applications like MATLAB or Octave excel. If your needs are efficient image algorithms under C and C++, OpenCV or VXL are good bets. Other libraries provide Bayesian inference or random number generators for a variety of probability distributions.

When interfacing a variety of sensors, a low-level language such as C is probably one of the best ways to develop a robust and efficient implementation. A problem arises only when a project requires performing many or all of these tasks under a single and sensible development framework, since each library declares its own data structures.

For example, an image grabbed by an OpenCV program cannot be directly sent to a MATLAB program that detects features. Development of mobile robotics software is one of those complex projects that require having at hand a variety of heterogeneous tools: a robot may capture an image from an IEEE1394 camera, extract features from it, read odometry information from wheel encoders through a serial port and then fuse all the data using a Kalman filter in matrix form. This contains tasks that range from low-level code (close to hardware), up to linear algebra. To face the development of such software, we have created Mobile Robot Programming Toolkit, or MRPT."

—Taken from 'Development of Scientific Applications with the MobileRobot Programming Toolkit', the MRPT reference book

Enhance the viewing perimeter

Computer Vision (CV) is one feature no robotics application is complete without. Understanding digital images and video, and translating data to process automatic visual tasks forms the core of CV, and this task of MRPT is largely supported by OpenCV. Images can be stored as grayscale or RGB bitmaps; tracking video feature lets you detect and track features of a video stream in real time.

Post better pictures on Instagram. A commonly disregarded fact is the radial and tangential distortion introduced by cameras. With interest in photography on the rise, everyone wants ideal projective camera images, and this is easily recovered by performing an undistortion of image pixels. The same applies even for video files, which are sequences of images, afterall.

You can calibrate camera sensors using tools like Program camera-calib, Complete Kinect calibration and Program stereo-calibgui, which help calibrate individual cameras, accurately reconstruct

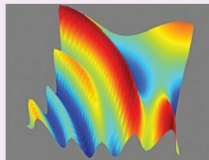


Fig. 2: The CMesh

depth and RGB information, and calibrate stereo cameras, respectively. Do you want to rectify stereo maps? Worry not, as MRPT lets you do that, too!

Locate and map on the same shot

Simultaneous Localisation and Mapping, commonly called SLAM in robotic circles, is defined by Wikipedia as the computational problem of constructing or updating a map of an unknown environment, while simultaneously keeping track of an agent's location within it. The way to do this is, of course, to use algorithms for a given time and environment.

The purpose defines the choice. MRPT provides you with a set of algorithms that can be readily